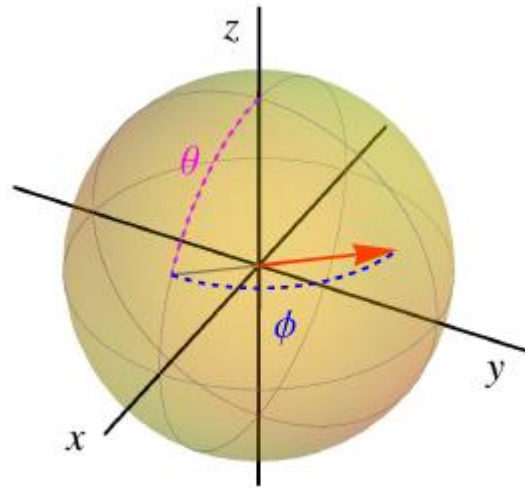


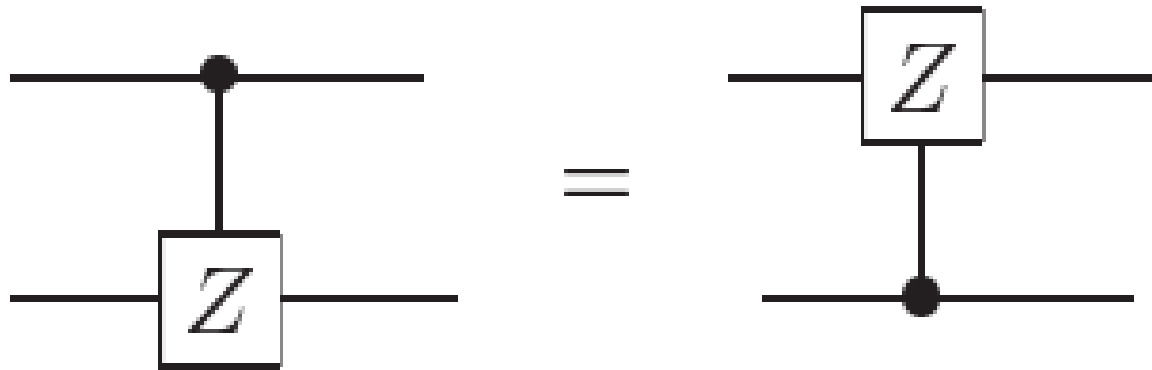
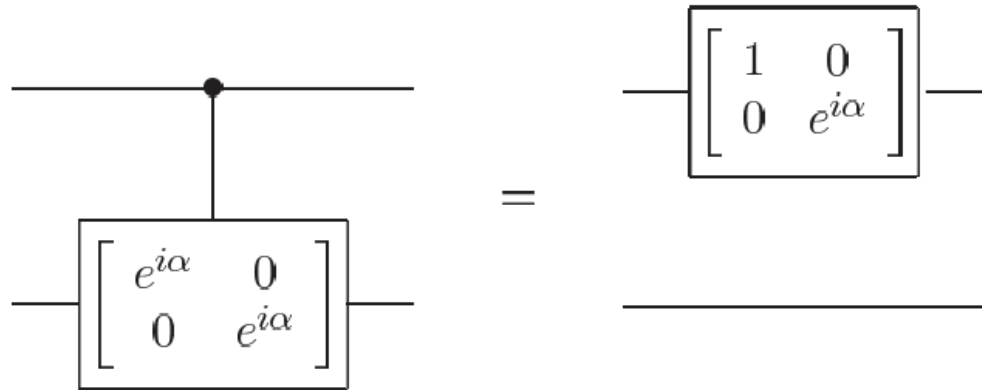
Quantum Computation

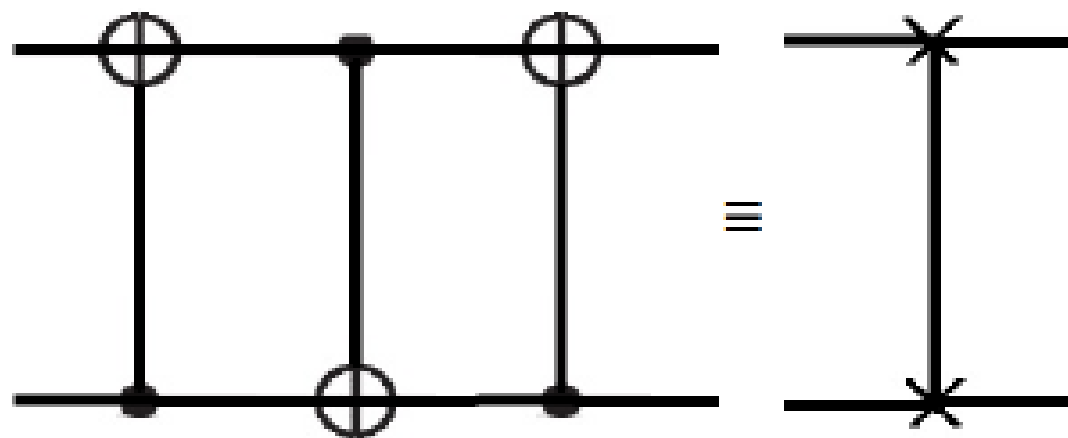
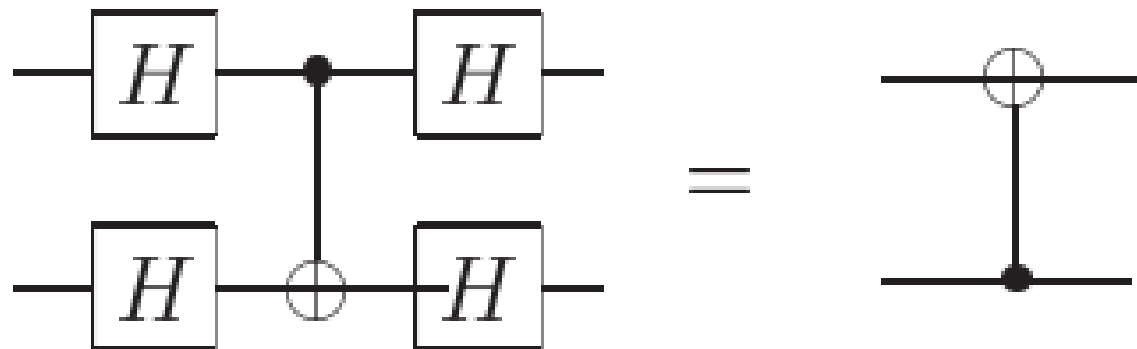
ELK336E –

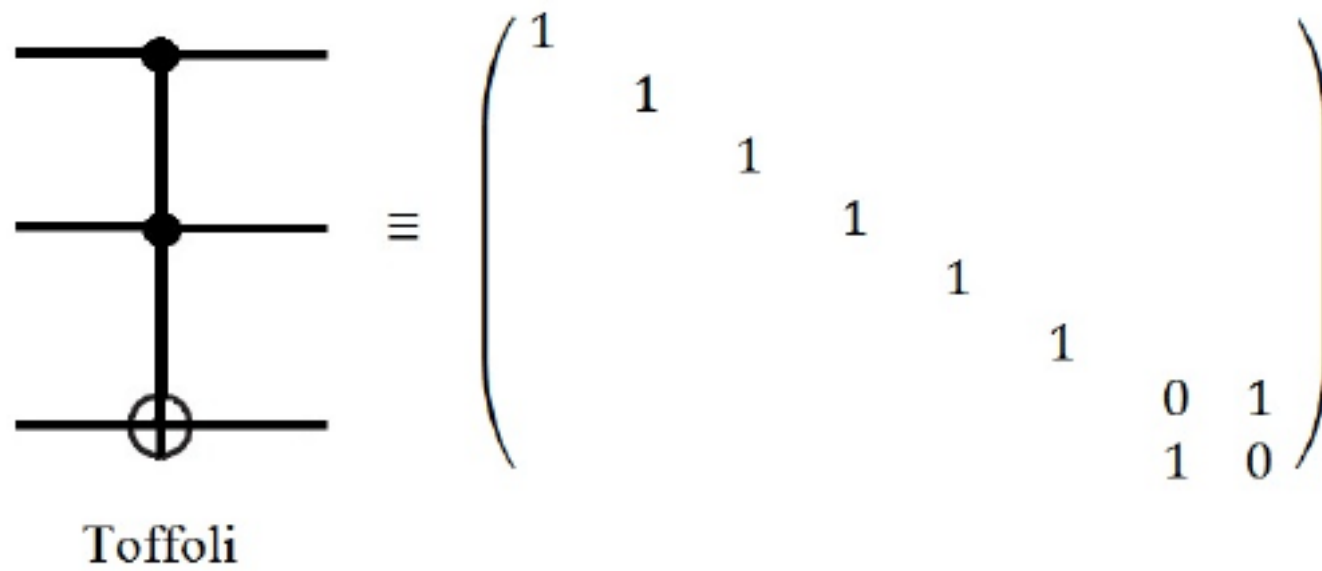
Week #6



Circuit identities:



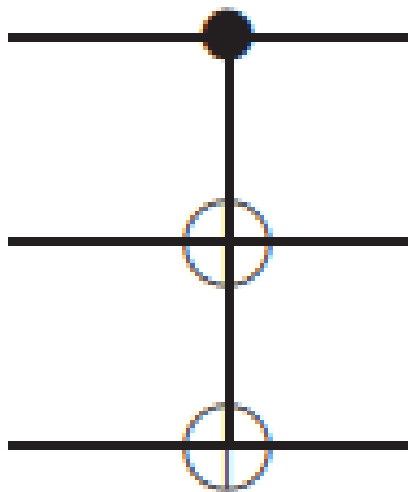




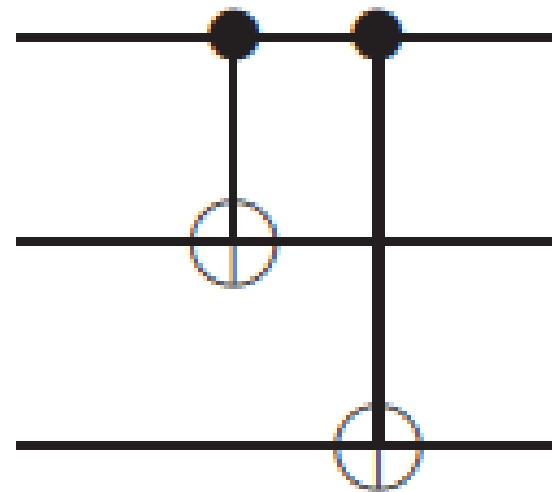
The diagram shows a Toffoli gate (CNOT with two controls) on three horizontal lines. The top two lines have control dots, and the bottom line has a target circle with a cross. To the right of the gate is an equivalence symbol followed by an 8x8 matrix. The matrix is block-diagonal, with a 6x6 identity block in the top-left and a 2x2 swap block in the bottom-right.

$$\text{Toffoli} \equiv \begin{pmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 0 & 1 \\ & & & & & & & 1 & 0 \end{pmatrix}$$

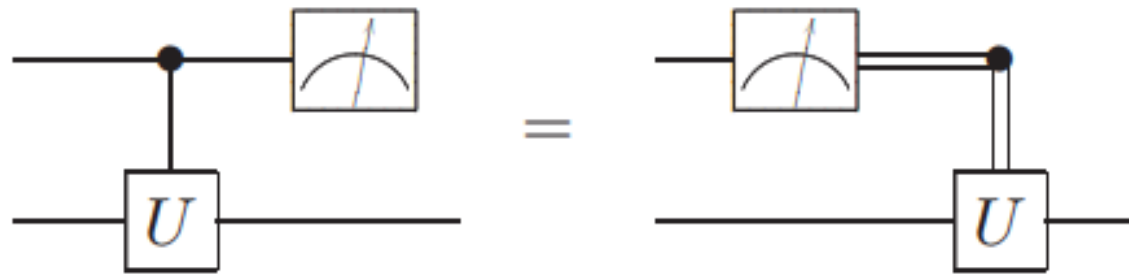
Toffoli

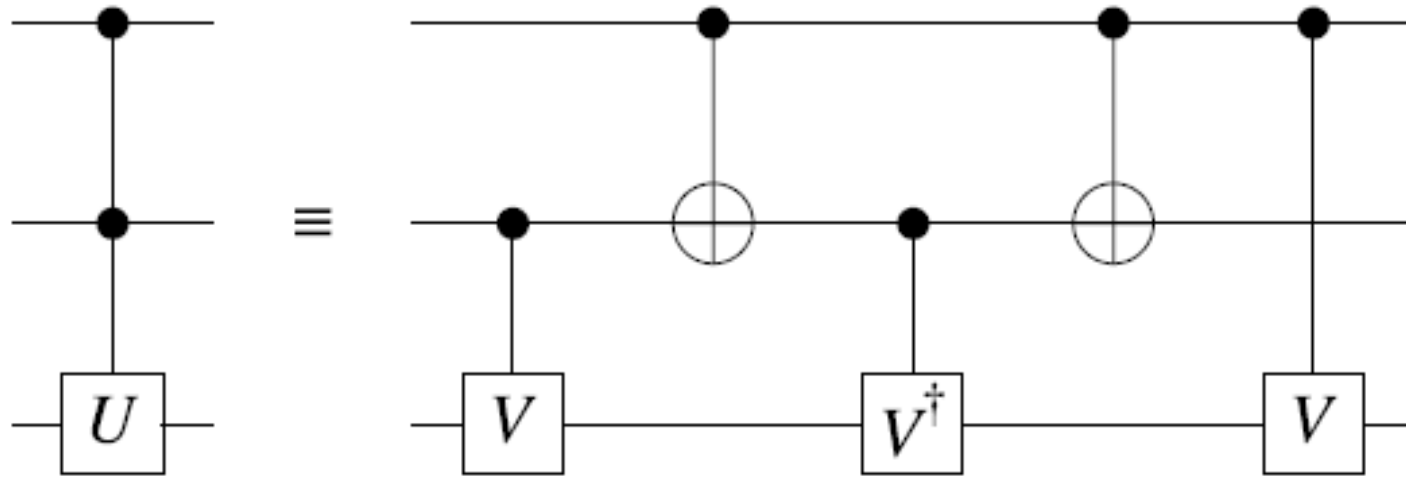


$\equiv$



Measurement commutes with controls





Quantum circuit for the controlled-controlled- U operation. Here V is any quantum gate such that $V^2 = U$.

EXAMPLE: Find the square root of Z and X .

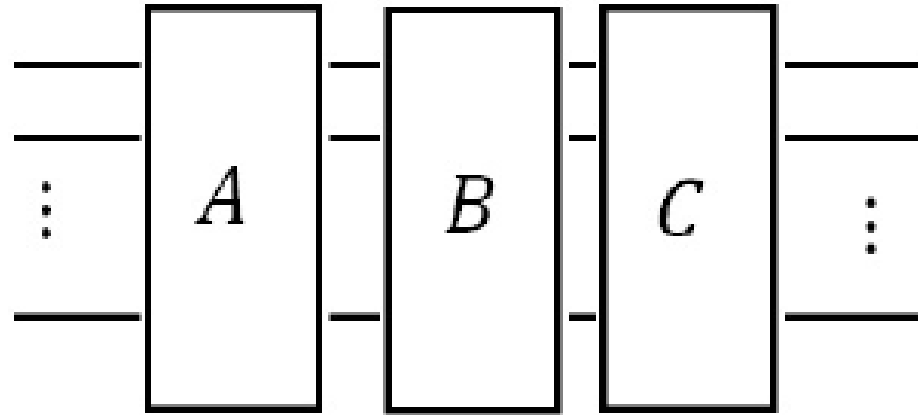
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$$

Where the $+1$ and the -1 on the right hand side are the eigenvalues.

$$\sqrt{Z} = \sqrt{+1}|0\rangle\langle 0| + \sqrt{-1}|1\rangle\langle 1| = (+1)|0\rangle\langle 0| + (i)|1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

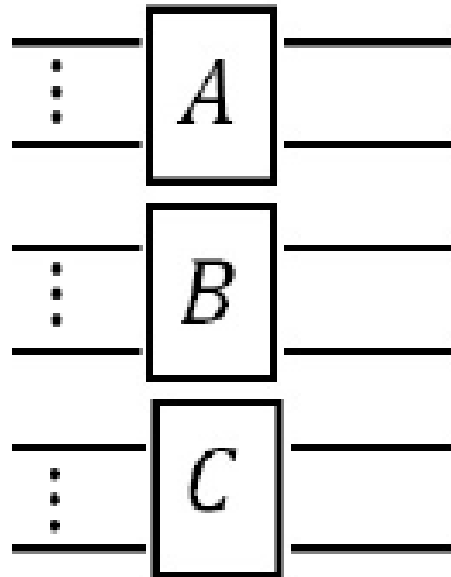
$$\sqrt{X} = (+1)|+\rangle\langle +| + (i)|-\rangle\langle -| = \begin{bmatrix} 0.5 + 0.5i & 0.5 - 0.5i \\ 0.5 - 0.5i & 0.5 + 0.5i \end{bmatrix}$$

Quantum circuits :

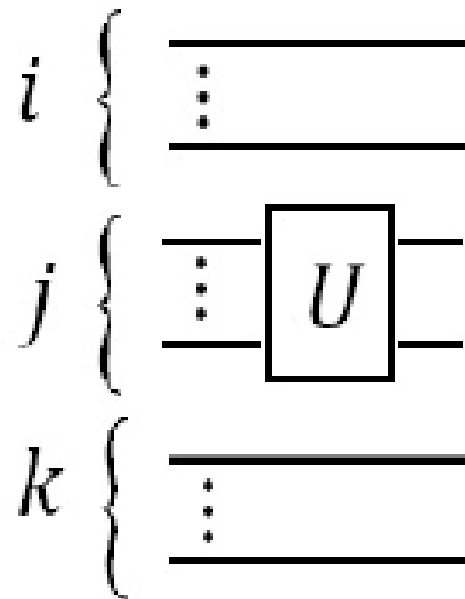


$$\equiv C \ B \ A$$

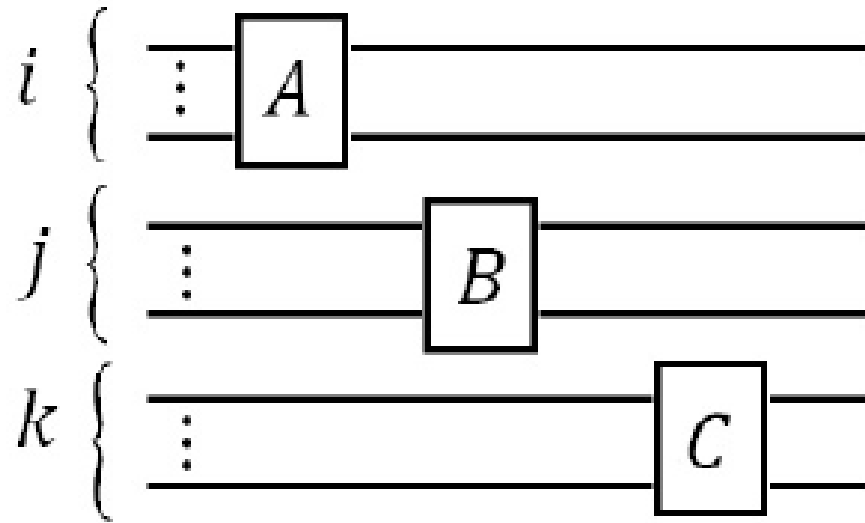
Time flows from left to right in the algorithms. But the implementation of the operators are from right to left.



$$\equiv A \otimes B \otimes C$$



$$\equiv (\mathbb{1}_{2^i} \otimes U \otimes \mathbb{1}_{2^k})$$

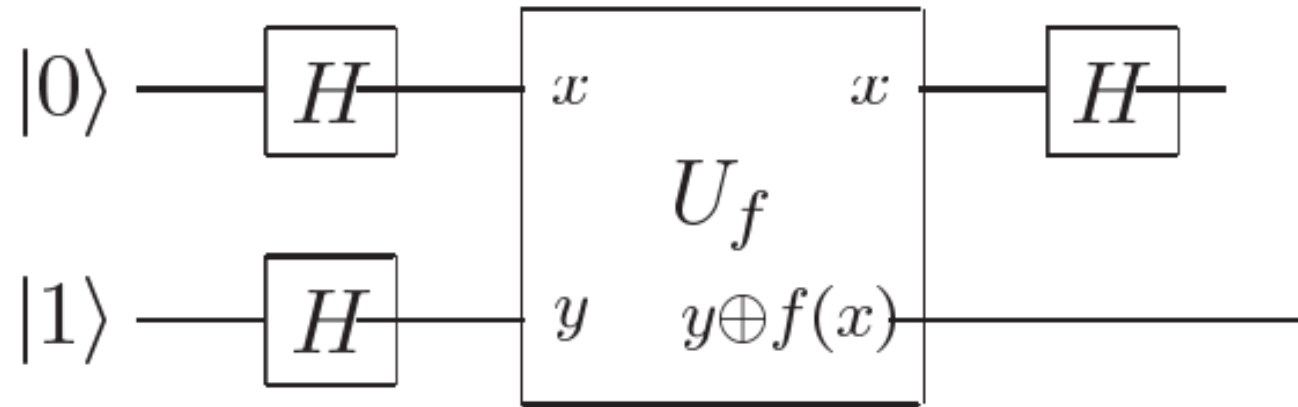


$$\equiv (A \otimes \mathbb{1}_{2^{j+k}}) \cdot (\mathbb{1}_{2^i} \otimes B \otimes \mathbb{1}_{2^k}) \cdot (\mathbb{1}_{2^{i+j}} \otimes C)$$

$$= A \otimes B \otimes C$$

Deutsch's algorithm

Deutsch's algorithm demonstrates how quantum circuits can outperform classical ones



Deutsch's algorithm finds whether a function is constant or balanced.
So, this is a decision algorithm.

Suppose $f(x) : \{0, 1\} \rightarrow \{0, 1\}$ is a function
with a one-bit domain and range.

if $f(0) = f(1)$, constant

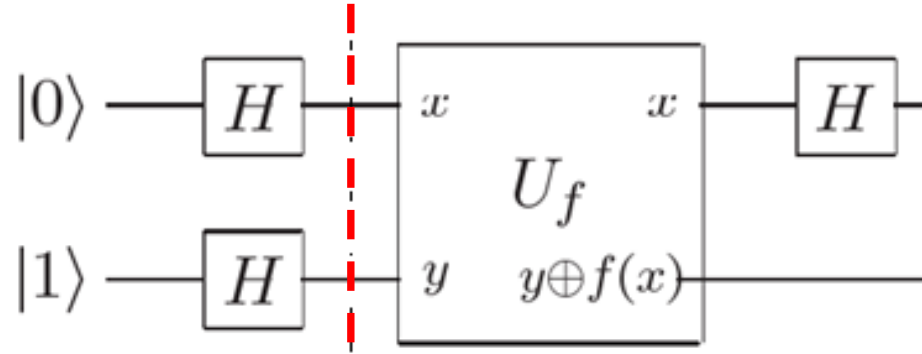
if $f(0) \neq f(1)$, balanced

Let's understand how the function works.

$$|x\rangle(|0\rangle - |1\rangle) \xrightarrow{f} |x\rangle(|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle)$$

x	$f(x)$	$ x\rangle(0 \oplus f(x)\rangle - 1 \oplus f(x)\rangle)$
0	0	$ 0\rangle(0\rangle - 1\rangle)$
0	1	$ 0\rangle(1\rangle - 0\rangle)$
1	0	$ 1\rangle(0\rangle - 1\rangle)$
1	1	$ 1\rangle(1\rangle - 0\rangle)$

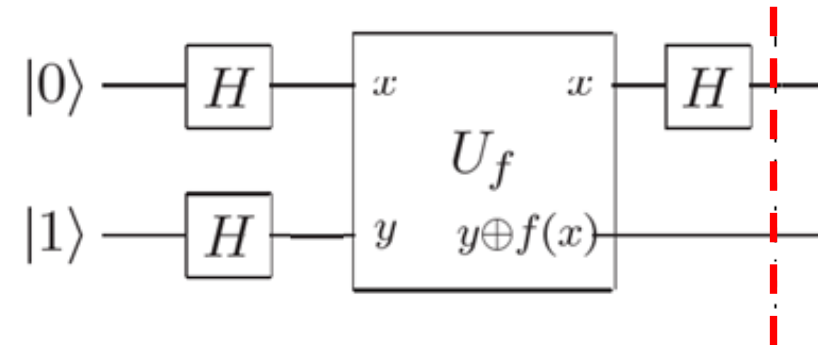
$$|x\rangle(|0\rangle - |1\rangle) \xrightarrow{f} (-1)^{f(x)} |x\rangle(|0\rangle - |1\rangle)$$



$$|01\rangle \xrightarrow{H \otimes H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\xrightarrow{f} \frac{1}{\sqrt{2}}((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

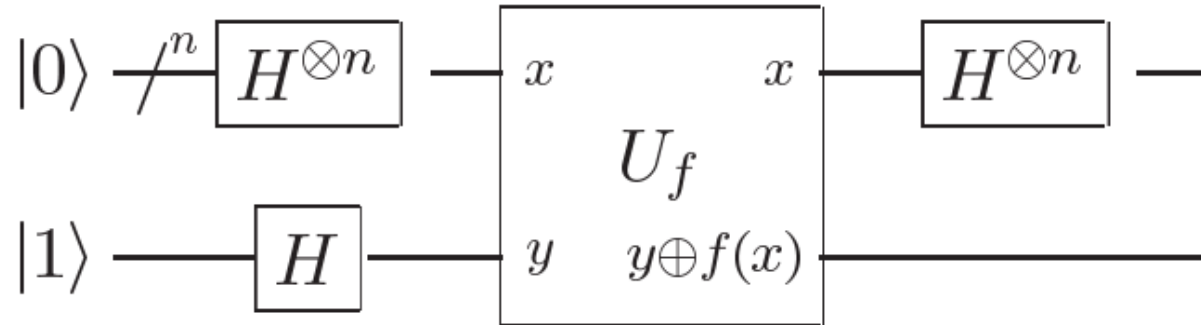
$$\xrightarrow{H \otimes \mathbb{1}} \left[\left(\frac{1}{2}(-1)^{f(0)} + \frac{1}{2}(-1)^{f(1)} \right) |0\rangle + \left(\frac{1}{2}(-1)^{f(0)} - \frac{1}{2}(-1)^{f(1)} \right) |1\rangle \right] \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$



According to this result, the algorithm decides whether the function is balanced or not just in one step.

(Quantum paralelism.)

The Deutsch-Jozsa algorithm



If we have many possibilities, the generalized form of D. algorithm can cope with this.

Entanglement :

A quantum state $|\Phi\rangle$ is said to be separable if it can be written in a tensor product form of two distinct quantum states. ;

$$|\Phi\rangle = |\phi_1\rangle \otimes |\phi_2\rangle : \text{Separable}$$

If the substates of $|\Phi\rangle$ can not be written in a tensor product form of its substates then this state is entangled. ;

$$|\Phi\rangle \neq |\phi_1\rangle \otimes |\phi_2\rangle : \text{Entangled}$$

For example $|00\rangle + |01\rangle$
is not an entangled state because:

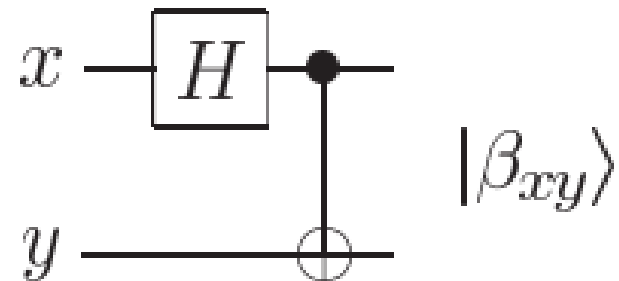
$$|00\rangle + |01\rangle = |0\rangle \otimes (|0\rangle + |1\rangle)$$

But $|01\rangle + |10\rangle$
is entangled.

Bell states

$$\begin{aligned}
 |\beta_{00}\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}}; & |\beta_{10}\rangle &= \frac{|00\rangle - |11\rangle}{\sqrt{2}}; \\
 |\beta_{01}\rangle &= \frac{|01\rangle + |10\rangle}{\sqrt{2}}; & |\beta_{11}\rangle &= \frac{|01\rangle - |10\rangle}{\sqrt{2}},
 \end{aligned}$$

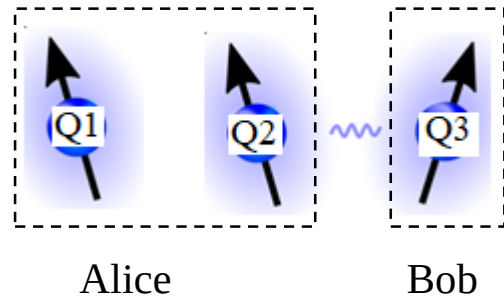
EPR states or EPR pairs,



In	Out
$ 00\rangle$	$(00\rangle + 11\rangle)/\sqrt{2} \equiv \beta_{00}\rangle$
$ 01\rangle$	$(01\rangle + 10\rangle)/\sqrt{2} \equiv \beta_{01}\rangle$
$ 10\rangle$	$(00\rangle - 11\rangle)/\sqrt{2} \equiv \beta_{10}\rangle$
$ 11\rangle$	$(01\rangle - 10\rangle)/\sqrt{2} \equiv \beta_{11}\rangle$

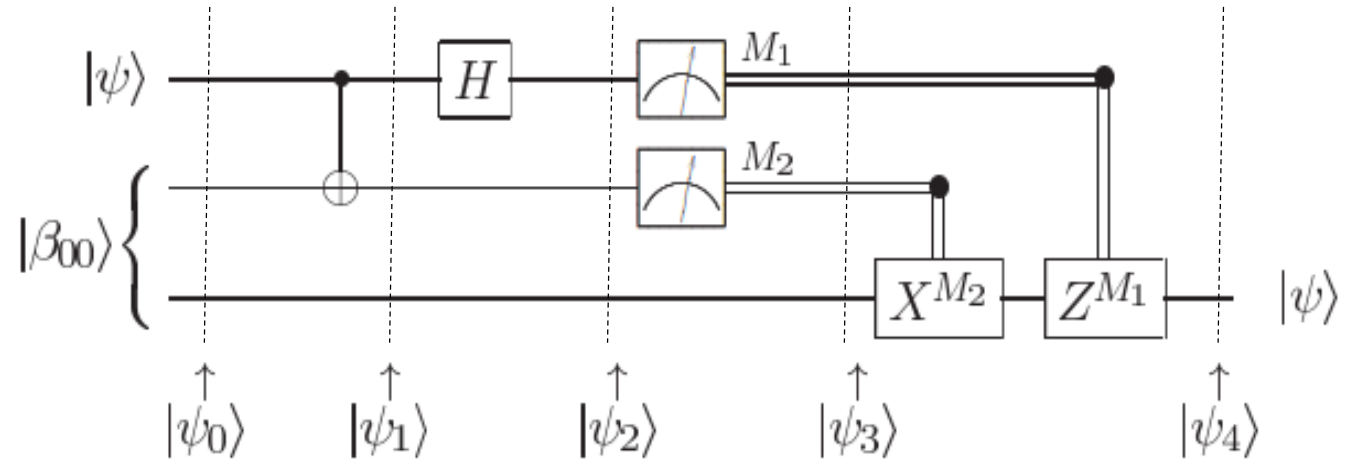
$$|\beta_{xy}\rangle \equiv \frac{|0, y\rangle + (-1)^x |1, \bar{y}\rangle}{\sqrt{2}},$$

Quantum teleportation:



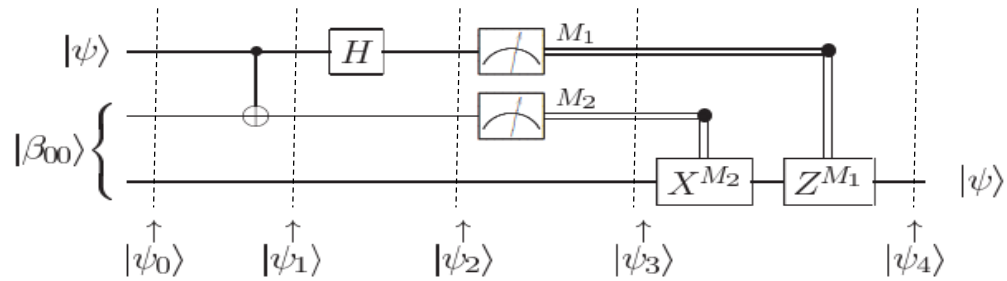
$|\psi\rangle$ is the qubit to be teleported.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



$$= \frac{1}{\sqrt{2}} \left[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle) \right]$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} \left[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle) \right]$$



$$|\psi_2\rangle = \frac{1}{2} \left[\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle) \right].$$

This state may be re-written in the following way, simply by regrouping terms:

$$|\psi_2\rangle = \frac{1}{2} \left[|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\alpha|1\rangle + \beta|0\rangle) + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (\alpha|1\rangle - \beta|0\rangle) \right].$$

Now Alice has to perform measurement to her qubits and send the information to BOB which state the system has been collapsed.

Result		Bob performs
$ 00\rangle$	then	1
$ 01\rangle$		X
$ 10\rangle$		Z
$ 11\rangle$		X and Z

For instance Alice has $|01\rangle$;
 this means that $|\psi_3\rangle = |01\rangle(\alpha|1\rangle + \beta|0\rangle)$
 Then BOB applies X operator to his qubit,
 this means that $|\psi_4\rangle = |01\rangle X(\alpha|1\rangle + \beta|0\rangle)$

$$|\psi_4\rangle = |01\rangle(\alpha|0\rangle + \beta|1\rangle)$$

$$|\psi_4\rangle = |01\rangle|\psi\rangle \quad \textbf{Teleportation is complete!}$$